

$$\frac{d^2y}{dx^2} - y = \cos x$$

$\implies$ order = 2, degree = 1

$$\left(\frac{d^2y}{dx^2}\right)^3 + \frac{dy}{dx} - 3y = \cos x$$

$\implies$ order = 2, degree = 3

Order of a Differential Equation

The highest order derivative present in a differential equation is known as its order.

Degree of a Differential Equation

The highest power of the highest order of derivative present in the differential equation is known as its degree provided that ALL the derivative terms present in the differential equation are made free from Fraction, Radical Sign, Trigonometry, Inverse Trigonometry, Logarithm and Exponents.

The order and degree of the differential

equation $\left(4 + \left(\frac{dy}{dx}\right)^2\right)^{\frac{2}{3}} = \frac{d^2y}{dx^2}$ are

A: Cubing both sides $\rightarrow$

$$\left[4 + (y_1)^2\right]^2 = (y_2)^3$$

$$\text{order} = 2$$

$$\text{degree} = 3$$

NOTE:

If ALL the derivative terms present in the differential Equation cannot be made free from fraction, radical sign, trigonometry, inverse trigonometry, logarithm and exponents then we say that the degree is not defined.

Ex:- $\sin \left(\frac{d^2 y}{dx^2} \right) = x + \left(\frac{dy}{dx} \right)$

$$\Rightarrow \frac{d^2 y}{dx^2} = \sin^{-1}(x + y_1)$$

order = 2, degree = N.D.

Steps to form Differential Equation for a given Family of Curves:

- **Step-I** → Differentiate the given family of curves as many times as the number of independent arbitrary constants.
- **Step-II** → Eliminate all the arbitrary constants from the given equation with the help of these equations to obtain the differential equation.

Form D. E. for $y^2 = 4ax \rightarrow$

a is the arbitrary constant

Differentiate once :

$$2y \frac{dy}{dx} = 2yy_1 = 4a$$

Divide $y^2 = 4ax$ with this $\Rightarrow$

$$\frac{y^2}{2yy_1} = \frac{4ax}{4a}$$

$$\Rightarrow \frac{y}{2y_1} = x$$

$$\Rightarrow y = 2x \frac{dy}{dx}$$

$$2x \frac{dy}{dx} = y$$

$$2 \int \frac{dy}{y} = \int \frac{dx}{x}$$

$$2 \ln y = \ln x + \ln c$$

$$\ln y^2 = \ln xc$$

$$y^2 = cx$$

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$\frac{y^2}{b^2} = \frac{a^2 - x^2}{a^2}$$

$$b^2 = \frac{a^2 y^2}{a^2 - x^2} \Rightarrow \frac{b^2}{a^2} = \frac{y^2}{a^2 - x^2}$$

Differentiate w.r.t x $\Rightarrow$

$$0 = \frac{(a^2 - x^2)2yy_1 - y^2(0 - 2x)}{(a^2 - x^2)^2}$$

$$2yy_1(a^2 - x^2) + 2xy^2 = 0$$

$$y_1(a^2 - x^2) + xy = 0$$

$$\Rightarrow a^2 - x^2 + \frac{xy}{y_1} = 0$$

$$a^2 = x^2 - \frac{xy}{y_1}$$

Differentiate w. r. t x again :

$$0 = 2x - \frac{y_1(xy_1 + y) - xyy_2}{y_1^2}$$

$$2xy_1^2 - (xy_1^2 + yy_1) + xyy_2 = 0$$

$$\mathbf{Ans} \rightarrow xy_1^2 - yy_1 + xyy_2 = 0$$

$$\mathbf{Q : } y = a \cos (x + b)$$

$$\mathbf{A : } \text{ Differentiate } \Rightarrow$$

$$y_1 = -a \sin(x + b)$$

$$y_2 = -a \cos(x + b)$$

$$y_2 = -y$$

$$\frac{d^2y}{dx^2} + y = 0$$

Q: Find the solution curve of the differential equation:

$$2y \frac{dy}{dx} + 3 = 5 \frac{dy}{dx}$$

provided that the curve passes through the point (0,1). Also find its vertex.

$$\mathbf{A} : 2y \frac{dy}{dx} - 5 \frac{dy}{dx} = -3$$

$$(2y - 5) \frac{dy}{dx} = -3$$

$$\int (2y - 5) dy = \int -3 dx$$

$$\int 2y dy - \int 5 dy = -3 \int dx$$

$$y^2 - 5y = -3x + c$$

(0,1) satisfies the equation $\rightarrow$

$$1 - 5 = (-3) \times 0 + c$$

$$c = -4$$

$$y^2 - 5y = -3x - 4$$

$$\left(y - \frac{5}{2}\right)^2 = -3x - 4 + \frac{25}{4}$$

$$\left(y - \frac{5}{2}\right)^2 = -3x + \frac{9}{4}$$

$$\left(y - \frac{5}{2}\right)^2 = -3\left(x - \frac{3}{4}\right)$$

$$Y^2 = -3X$$

$$\text{Vertex: } \left(\frac{3}{4}, \frac{5}{2}\right)$$

Homogenous Functions – Case (1)

They are functions where all terms have the same degree.

$$f(x, y) = ax^2 + by^2 + 2hxy$$

$$f(x, y) = ax^3 + by^3 + cxy^2 + dx^2y$$

If $f(\lambda x, \lambda y) = \lambda^n f(x, y) \rightarrow$
 n is called the degree and the
given equation is homogenous.

$$\text{E.g. : } f(x, y) = \frac{3y - 7x}{3x + y}$$

$$\begin{aligned} f(\lambda x, \lambda y) &= \frac{3\lambda y - 7\lambda x}{3\lambda x + \lambda y} \\ &= \frac{\lambda(3y - 7x)}{\lambda(3x + y)} = f(x, y) \end{aligned}$$

(zero degree homogenous equation)

**Homogenous D.E. can be solved by
using the following substitution $\rightarrow$**

$$\text{When : } \frac{dy}{dx} = f(x, y)$$

$$\text{Use : } y = Vx$$

Q: $y = y(x)$ is the solution of the following differential equation:

$$2x^2 \frac{dy}{dx} - 2xy + 3y^2 = 0$$

$$\text{and } y(e) = \frac{e}{3}$$

then find the value of $y(1)$.

Answer :

$$\frac{dy}{dx} = \frac{2xy - 3y^2}{2x^2}$$

Let $y = Vx$ and differentiate :

$$\frac{dy}{dx} = V \frac{dx}{dx} + x \frac{dV}{dx} = V + x \frac{dV}{dx}$$

$$V + x \frac{dV}{dx} = \frac{2Vx^2 - 3V^2x^2}{2x^2}$$

$$V + x \frac{dV}{dx} = \frac{2V - 3V^2}{2}$$

$$V + x \frac{dV}{dx} = V - \frac{3V^2}{2}$$

$$x \frac{dV}{dx} = -\frac{3V^2}{2}$$

$$\int \frac{dV}{V^2} = -\frac{3}{2} \int \frac{dx}{x}$$

$$-\frac{1}{V} = -\frac{3}{2} \ln x - C$$

$$\frac{1}{V} = \frac{3}{2} \ln x + C$$

$$\frac{x}{y} = \frac{3}{2} \ln x + C \Rightarrow y(e) = e/3$$

$$\frac{e}{e/3} = \frac{3}{2} + C$$

$$3 - \frac{3}{2} = C$$

$$C = \frac{3}{2}$$

$$\frac{x}{y} = \frac{3}{2} (\ln x + 1)$$

$$x = 1$$

$$\frac{1}{y} = \frac{3}{2} \times 1$$

$$y = \frac{2}{3}$$

Homogenous Functions – Case (2)

$$\text{When : } \frac{dx}{dy} = f(x, y)$$

$$\text{Use : } x = Vy$$

Linear Differential Equations – Case (1)

They are of the following form:

$$\frac{dy}{dx} + (P(x) \times y) = Q(x)$$

To solve such questions, we require a quantity called Integrating Factor (IF) which is given by $\rightarrow$

$$\mathbf{I.F} = \mathbf{e}^{\int \mathbf{P(x)} \, \mathbf{dx}}$$

The final answer is obtained with the help of the integrating factor $\rightarrow$

$$\mathbf{y \, (IF) = \int Q(x) \, (IF) \, dx}$$

$$\frac{dy}{dx} + \frac{y}{x} = x^2$$

$$\frac{dy}{dx} + \frac{1}{x}y = x^2$$

$$\frac{1}{x} = P(x)$$

$$x^2 = Q(x)$$

$$\begin{aligned}\text{I.F.} &= e^{\int \frac{1}{x} dx} \\ &= e^{\ln x} \\ &= x\end{aligned}$$

$$\implies \text{I.F.} = x$$

$$y \cdot x = \int x^2 \cdot x dx$$

$$y \cdot x = \int x^3 dx$$

$$y \cdot x = \frac{x^4}{4} + C$$

Linear Differential Equations – Case (2)

They are of the following form:

$$\frac{dx}{dy} + P(y) x = Q(y)$$

$$\mathbf{I. F.} = \mathbf{e}^{\int P(y) dy}$$

$$x \cdot (\text{I.F.}) = \int Q(y) (\text{I.F.}) dy$$

Equations Reducible to Homogenous D.E.

$$\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$$

Case (1) → If the given lines are parallel:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

then substitute →

$$a_1x + b_1y = t$$

Case (2) → When:

$$a_2 + b_1 = 0$$

then a simple cross multiplication and
substitution of the following kind →

$$d(xy) \text{ for } x dy + y dx$$

along with integrating term by
term yields the result easily.

Case (3) → If neither of the above
cases is true, then the equation is
reducible to homogenous form.

Case (1) Example →

**Q: Find the solution of the following D.E.
provided that $y(0) = 3$**

$$(2x + 3y - 2) dx + (4x + 6y - 7) dy = 0$$

Answer →

$$(2x + 3y - 2) + (4x + 6y - 7) \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x + 3y - 2}{4x + 6y - 7}$$

$$\implies \text{Let } 2x + 3y = t$$

$$2 + 3 \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{\frac{dt}{dx} - 2}{3}$$

$$\frac{\frac{dt}{dx} - 2}{3} = \frac{-(t - 2)}{2t - 7}$$

$$\frac{dt}{dx} - 2 = \frac{-3(t - 2)}{2t - 7}$$

$$\frac{dt}{dx} = 2 - \frac{3(t-2)}{2t-7}$$

$$\frac{dt}{dx} = \frac{2(2t-7) - 3(t-2)}{2t-7}$$

$$\frac{dt}{dx} = \frac{4t - 14 - 3t + 6}{2t-7}$$

$$\frac{dt}{dx} = \frac{t-8}{2t-7}$$

$$\frac{2t-7}{t-8} dt = dx$$

Take :

$$\frac{2t-7}{t-8} = \frac{2(t-8) + 16 - 7}{t-8}$$

$$= \frac{2(t-8) + 9}{t-8} = \frac{2(t-8)}{t-8} + \frac{9}{t-8}$$

$$= 2 + \frac{9}{t-8}$$

$$\left(2 + \frac{9}{t-8}\right) dt = dx$$

$$\int \left(2 + \frac{9}{t-8}\right) dt = \int dx$$

$$2t + 9 \ln |t-8| = x + C$$

$$2(2x + 3y) + 9 \ln |2x + 3y - 8| = x + C$$

$$4x + 6y + 9 \ln |2x + 3y - 8| = x + C$$

$$3x + 6y + 9 \ln |2x + 3y - 8| = K$$

$$\text{Given condition: } y(0) = 3$$

$$3(0) + 6(3) + 9 \ln |2(0) + 3(3) - 8| = K$$

$$0 + 18 + 9 \ln |9 - 8| = K$$

$$18 + 9 \ln(1) = K$$

$$18 + 0 = K$$

$$K = 18$$

$$\text{Ans. : } 3x + 6y + 9 \ln |2x + 3y - 8| = 18$$

Case (2) Example →

Solve : $\frac{dy}{dx} = \frac{x - 2y + 1}{2x + 2y + 3}$

Answer : Cross multiply →

$$2x \, dy + 2y \, dy + 3 \, dy$$

$$= x \, dx - 2y \, dx + dx$$

$$2(x \, dy + y \, dx) + 2y \, dy + 3 \, dy$$

$$= x \, dx + dx$$

$$2 \int d(xy) + \int 2y \, dy + \int 3 \, dy$$

$$= \int x \, dx + \int dx$$

$$2xy + y^2 + 3y = \frac{x^2}{2} + x + C$$

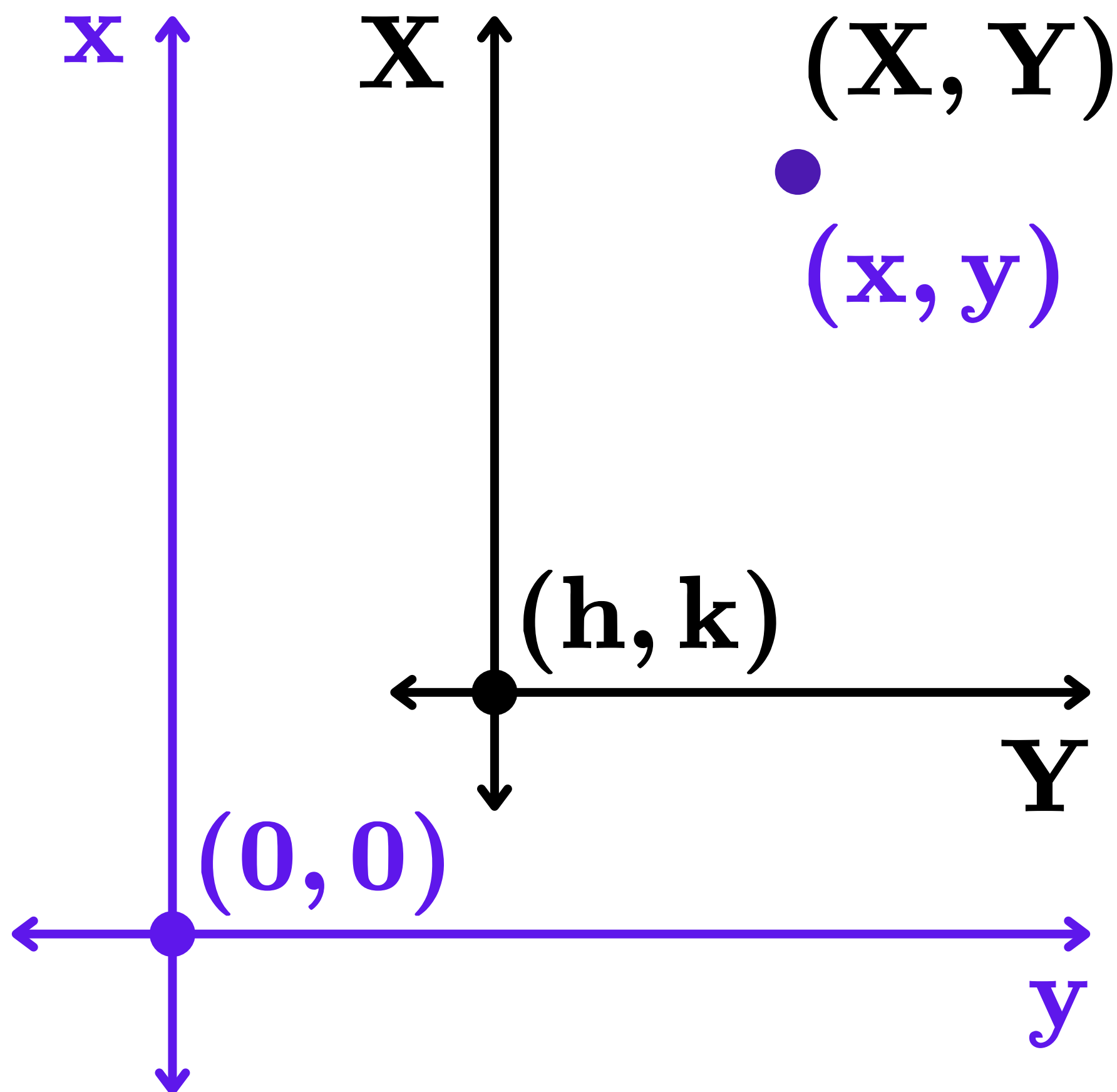
Case (3) Example →

For problems from Case (3), we need to shift the origin to the point where both the following lines intersect →

$$a_1x + b_1y + c_1 = 0$$

$$a_2x + b_2y + c_2 = 0$$

Let their point of intersection be: (h, k)



$$\text{Substitute : } X = x - h$$

$$Y = y - k$$

$$\frac{dY}{dX} = \frac{a_1X + b_1Y}{a_2X + b_2Y} \implies Y = VX$$

Solve : $\frac{dy}{dx} = \frac{y - x + 1}{y + x + 5}$

Answer :

$$-x + y + 1 = 0$$

$$x + y + 5 = 0$$

$$(\mathbf{h}, \mathbf{k}) = (-2, -3)$$

$$X = x - h \implies X = x + 2$$

$$Y = y - k \implies Y = y + 3$$

$$\frac{dy}{dx} = \frac{-x + y + 1}{x + y + 5}$$

$$\frac{dY}{dX} = \frac{-X + Y}{X + Y} \implies Y = VX$$

$$\implies V + X \frac{dV}{dX} = \frac{-X + VX}{X + VX}$$

$$X \frac{dV}{dX} = \frac{V - 1}{V + 1} - V$$

$$X \frac{dV}{dX} = \frac{V - 1 - V^2 - V}{V + 1}$$

$$X \frac{dV}{dX} = - \frac{1 + V^2}{V + 1}$$

$$\int \frac{(V + 1) dV}{V^2 + 1} = - \int \frac{dX}{X}$$

$$\frac{1}{2} \int \frac{2V dV}{V^2 + 1} + \int \frac{dV}{V^2 + 1} + \int \frac{dX}{X} = 0$$

$$\frac{1}{2} \ln(V^2 + 1) + \tan^{-1} V + \ln |X| = C$$

$$\frac{1}{2} \ln \left[\left(\frac{Y}{X} \right)^2 + 1 \right] + \tan^{-1} \frac{Y}{X} + \ln |X| = C$$

$$\frac{1}{2} \ln \left(\frac{Y^2}{X^2} + 1 \right) + \tan^{-1} \left(\frac{Y}{X} \right) + \ln X = C$$

$$\frac{1}{2} \ln \left(\frac{Y^2 + X^2}{X^2} \right) + \tan^{-1} \left(\frac{Y}{X} \right) + \ln X = C$$

$$\frac{1}{2} \ln(Y^2 + X^2) - \ln(X) + \tan^{-1} \left(\frac{Y}{X} \right) + \ln X = C$$

$$\frac{1}{2} \ln (X^2 + Y^2) + \tan^{-1} \left(\frac{Y}{X} \right) = C$$

$$\frac{1}{2} \ln((x + 2)^2 + (y + 3)^2) + \tan^{-1} \left(\frac{y + 3}{x + 2} \right) = C$$

$$\frac{1}{2} \ln(x^2 + 4x + 4 + y^2 + 6y + 9) + \tan^{-1} \left(\frac{y + 3}{x + 2} \right) = C$$

$$\frac{1}{2} \ln (x^2 + y^2 + 4x + 6y + 13) + \tan^{-1} \left(\frac{y + 3}{x + 2} \right) = C$$